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- During the attendance check a sticker containing a unique code will be put on this exam.
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## Statistics for Business Administration

**Exam:** MA9712 / Practice Exam C

**Date:** Tuesday 6th October, 2026

**Examiner:** Prof. Mathias Drton

**Time:** 09:30 – 11:00

### Working instructions

- This exam consists of pages with a total of 9 problems.  
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is **52 credits**.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one scientific pocket calculator
  - the cheat sheet, which is contained in this exam on page 16
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from \_\_\_\_\_ to \_\_\_\_\_

Early submission at \_\_\_\_\_

**Problem 1**

7.5 credits

A city gym surveyed all 550 of its members about their satisfaction with a newly introduced group-class schedule. Each member answered *Positive*, *Neutral*, or *Negative*, and each member's *membership tier* (Basic, Standard, Premium) was recorded. The results are summarised in the contingency table below.

|          | Positive | Neutral | Negative | Total |
|----------|----------|---------|----------|-------|
| Basic    | 100      | 90      | 60       | 250   |
| Standard | 110      | 55      | 35       | 200   |
| Premium  | 65       | 20      | 15       | 100   |
| Total    | 275      | 165     | 110      | 550   |

- Identify the response variable and the explanatory variable in this study.
- Is this study observational or experimental? Can the gym conclude that the membership tier *causes* a member's satisfaction level? Does the result generalise, and if so, to which population?
- Which graphical display is appropriate to compare the distribution of satisfaction across the three tiers? Justify briefly.
- Compute the probability that a randomly chosen member is a Premium member *and* answered Positive.
- Show that  $\mathbb{P}(\text{Negative}) = 0.200$ .
- Suppose 10 members are drawn independently at random (with replacement) from a large population in which the proportion of Negative responses equals  $\mathbb{P}(\text{Negative})$  from part e). Let  $X$  be the number who answer Negative. Compute  $\mathbb{P}(X = 3)$ . You may use:

```
| dbinom(3, size = 10, prob = 0.2) ## [1] 0.2013266
| pbinom(3, size = 10, prob = 0.2) ## [1] 0.8791261
```

**Problem 2**

6 credits

A meteorologist records the total monthly rainfall (in mm) at a weather station for a random sample of  $n = 40$  months. The sample mean is  $\bar{x} = 85$  mm with sample standard deviation  $s = 24$  mm. A boxplot of the 40 values is roughly symmetric with no outliers; its lower and upper quartiles are  $Q_1 = 68$  mm and  $Q_3 = 102$  mm.

- Read the interquartile range (IQR) of the monthly rainfall from the information given.
  - State the conditions required for the sampling distribution of  $\bar{x}$  to be approximately normal (Central Limit Theorem), and argue whether they are met here.
  - Construct the 95% confidence interval for the true mean monthly rainfall  $\mu$ . You may use:
- ```
| qnorm(0.975) ## [1] 1.959964
```
- Interpret your confidence interval from part c) in the context of the study.

**Problem 3**

4 credits

A hardware factory produces bolts on three machines,  $M_1$ ,  $M_2$  and  $M_3$ , which account for 25%, 35% and 40% of total output, respectively. The proportion of defective bolts is 5% for  $M_1$ , 3% for  $M_2$  and 2% for  $M_3$ . One bolt is drawn at random from the combined output.

- Define suitable events and describe the corresponding probability tree (branches and their probabilities).
- The drawn bolt is found to be defective. Compute the probability that it was produced by machine  $M_1$ . Round to three decimal places.

**Problem 4**

5 credits

An online retailer claims that at most 4% of all orders are returned. A consumer group inspects a random sample of  $n = 200$  orders and finds that 14 of them were returned. Test, at significance level  $\alpha = 0.05$ , whether the data provide convincing evidence that the true return rate exceeds 4%.

- State the null and alternative hypotheses in terms of the true return proportion  $p$ .
- Let  $X \sim \text{Bin}(200, 0.04)$  be the number of returned orders under  $H_0$ . Compute the  $p$ -value of the test. You may use:

```
pbinom(13, size = 200, prob = 0.04)          ## [1] 0.9687882
pbinom(13, size = 200, prob = 0.04, lower.tail = FALSE) ## [1] 0.0312118
```

- State your decision and conclusion in context.

**Problem 5**

4 credits

Decide whether each of the following statements is **true** or **false** and *justify* your answer.

*Remark: Only correct justifications are awarded with points, not the decision (true/false).*

- A Type I error occurs when we fail to reject a null hypothesis that is in fact false.
- If a 95% confidence interval for a parameter does not contain the value  $\theta_0$ , then the two-sided test of  $H_0 : \theta = \theta_0$  at level  $\alpha = 0.05$  rejects  $H_0$ .
- Holding everything else fixed, increasing the sample size  $n$  increases the standard error of  $\hat{p}$  and therefore decreases the absolute value of the test statistic  $Z = (\hat{p} - p_0)/\text{SE}_0$ .
- “We are 95% confident that  $\mu$  lies in the interval” means that there is a 95% probability that this particular computed interval contains  $\mu$ .

**Problem 6**

4 credits

In a clinical trial, 300 patients were randomly assigned to receive Drug A, Drug B, or a placebo (100 per group), and each patient's outcome was recorded as *Improved* or *Not improved*. The observed counts are:

|         | Improved | Not improved | Total |
|---------|----------|--------------|-------|
| Drug A  | 60       | 40           | 100   |
| Drug B  | 55       | 45           | 100   |
| Placebo | 35       | 65           | 100   |
| Total   | 150      | 150          | 300   |

- a) State the null and alternative hypotheses of a  $\chi^2$  test of independence between treatment group and outcome.
- b) Compute the expected count in each cell and verify that the condition for the  $\chi^2$  approximation is satisfied.
- c) The test statistic equals  $\chi^2 = 14$ . State the degrees of freedom and bracket the  $p$ -value. You may use:

```
qchisq(0.95, df = 2) ## [1] 5.991465
qchisq(0.99, df = 2) ## [1] 9.210340
qchisq(0.999, df = 2) ## [1] 13.815511
```

Conclude in context at  $\alpha = 0.05$ .

## Problem 7

9 credits

A retail chain tested four store layouts (W, X, Y, Z) by applying each layout to 5 randomly chosen stores and recording the average daily sales (in hundreds of euros) over one week. The group summary is:

```
# A tibble: 4 x 3
  layout      n mean
  <chr> <int> <dbl>
1 W         5   200
2 X         5   210
3 Y         5   190
4 Z         5   220
# grand mean = 205
```

- a) State the null and alternative hypotheses of the one-way ANOVA.
- b) State the three conditions required for ANOVA and name what you would check for each.
- c) Complete the missing entries (i)–(v) in the ANOVA table below, given that the sum of squares between groups is  $SSG = 2500$  and the sum of squares within groups (residual) is  $SSE = 1600$ .

|           | Df   | Sum Sq | Mean Sq | F value |
|-----------|------|--------|---------|---------|
| layout    | (i)  | 2500   | (iii)   | (v)     |
| Residuals | (ii) | 1600   | (iv)    |         |

- d) Carry out the  $F$ -test at  $\alpha = 0.05$  and conclude in context. You may use:

```
qf(0.95, df1 = 3, df2 = 16) ## [1] 3.238872
qf(0.99, df1 = 3, df2 = 16) ## [1] 5.292214
```

- e) What percentage of the total variability in daily sales is explained by the store layout?
- f) If all pairwise comparisons of the four layouts are tested afterwards using a Bonferroni correction at family-wise level  $\alpha = 0.05$ , what corrected significance level  $\alpha^*$  should be used for each individual comparison?

**Problem 8**

8.5 credits

An analyst models the fuel efficiency (**mpg**, miles per gallon) of  $n = 32$  cars using their weight (**wt**, in 1000 lbs), gross horsepower (**hp**), and number of cylinders (**cyl**). The fitted `lm()` summary is:

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  40.000      3.000   13.33  3.1e-13 ***
wt          -3.200      0.600   -5.33  1.2e-05 ***
hp           -0.020      0.007   -2.86  0.0080 **
cyl          -0.800      0.550   -1.45  0.1570

Residual standard error: 2.5 on 28 degrees of freedom

```

- Write down the fitted regression equation.
- Interpret the slope coefficients of **wt** and of **hp** in context.
- State the null and alternative hypotheses for testing whether **hp** has an effect on **mpg**, and give your decision at  $\alpha = 0.05$ .
- Compute the adjusted coefficient of determination  $R^2_{\text{adj}}$ . You may use:
 

```
| var(mpg)  ## [1] 36
```
- A backward elimination based on AIC is performed. Which single predictor is most likely to be dropped first, and why?

**Problem 9**

5 credits

A university models whether an applicant is admitted (**admit** = 1) or rejected (**admit** = 0) from the applicant's grade point average (**gpa**), **gender** (**male** = 1, **female** = 0), and standardised **test\_score**, including a **gpa** × **gender** interaction. The fitted `glm(family = binomial)` output is:

```

Call:
glm(formula = admit ~ gpa * gender + test_score,
    family = _____, data = applicants)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -6.0000    1.4000   -4.29  1.8e-05 ***
gpa           1.5000    0.4000    3.75  0.0002 ***
gender        -2.0000    0.9000   -2.22  0.0264 *
test_score     0.0100    0.0030    3.33  0.0009 ***
gpa:gender      0.8000    0.3500    2.29  0.0221 *

```

- Fill in the blank in the `glm()` call above (the argument to **family**).
- Write the fitted model (the estimated *logit*) for a *female* applicant, simplified.
- Interpret the coefficient of **test\_score** as an odds ratio.
- Using the classification table below (with a threshold of 0.5), compute the sensitivity and the specificity of the classifier. The fitted model achieves an  $\text{AUC} = 0.877$ ; state in one sentence what this value measures.

|              | Actual admit | Actual reject |
|--------------|--------------|---------------|
| Pred. admit  | 40           | 15            |
| Pred. reject | 10           | 85            |

**Cheat sheet**

**Exploring data.**  $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ ,  $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$ ,  $s = \sqrt{s^2}$ , IQR =  $Q_3 - Q_1$ , fences  $Q_1 - 1.5 \text{ IQR}$ ,  $Q_3 + 1.5 \text{ IQR}$ .

$$r_{(x,y),n} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x s_y}$$

**Probability.**  $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$ ;  $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$  if independent;  $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ ;  
 $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ .

$$\mathbb{E}(X) = \sum_i x_i \mathbb{P}(X = x_i) \text{ (discrete)}, \quad \mathbb{E}(X) = \int x f(x) dx \text{ (cont.)},$$

$$\text{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2, \quad \text{SD}(X) = \sqrt{\text{Var}(X)}, \quad \text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y), \quad Z = \frac{X - \mathbb{E}(X)}{\sqrt{\text{Var}(X)}}.$$

**Distributions.**

$$\text{Geometric: } \mathbb{P}(X = x) = (1-p)^{x-1}p, \quad \mathbb{E}(X) = \frac{1}{p}, \quad \text{SD}(X) = \sqrt{\frac{1-p}{p^2}}.$$

$$\text{Binomial: } \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \mathbb{E}(X) = np, \quad \text{SD}(X) = \sqrt{np(1-p)}.$$

$$\text{Neg. Binomial: } \mathbb{P}(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x \geq k. \quad \text{Poisson: } \mathbb{P}(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \mathbb{E}(X) = \lambda, \quad \text{SD}(X) = \sqrt{\lambda}.$$

**Inference — categorical.**

$$1 \text{ prop. CI: } \hat{p} \pm z^* \text{SE}(\hat{p}), \quad \text{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}; \quad \text{test: } Z = \frac{\hat{p} - p_0}{\text{SE}_0}, \quad \text{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}}.$$

$$2 \text{ prop. CI: } (\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}; \quad \text{test: } \text{SE}_0 = \sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}, \quad \hat{p}_{\text{pool}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}.$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \frac{(\text{row } i)(\text{col } j)}{\text{total}}, \quad df = k - 1 \text{ (GOF) or } (r - 1)(c - 1).$$

**Inference — numerical.**

$$1\text{-sample } t: \bar{x} \pm t_{df}^* \text{SE}(\bar{x}), \quad T_{df} = \frac{\bar{x} - \mu_0}{\text{SE}(\bar{x})}, \quad \text{SE}(\bar{x}) = \frac{s}{\sqrt{n}}, \quad df = n - 1.$$

$$2\text{-sample } t: (\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \text{SE}, \quad T_{df} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\text{SE}}, \quad \text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \quad df = \min(n_1 - 1, n_2 - 1).$$

$$\text{Sample size: two-sample } n \geq \frac{2\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample } n \geq \frac{\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2, \quad \alpha^* \in \{\alpha, \alpha/2\}.$$

$$\text{ANOVA \& regression. } F = \frac{\text{MSG}}{\text{MSE}}, \quad df_1 = k - 1, \quad df_2 = n - k.$$

$$\text{SSG} = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2, \quad \text{SST} = \sum_{j=1}^n (y_j - \bar{y})^2, \quad \text{SSE} = \text{SST} - \text{SSG}.$$

$$\text{Post-hoc } T_{df} = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{\text{MSE}/n_i + \text{MSE}/n_j}}, \quad df = n - k; \quad \text{Bonferroni } \alpha^* = \alpha/K, \quad p_{\text{adj},i} = K p_i, \quad K = \binom{k}{2}.$$

$$\hat{y} = b_0 + b_1 x, \quad b_1 = r_{(x,y),n} \frac{s_y}{s_x}, \quad \hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}, \quad T_{df} = \frac{b_1 - \beta_{1,0}}{\text{SE}_{b_1}}, \quad df = n - 2.$$

$$R^2 = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}, \quad R_{\text{adj}}^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum (y_i - \bar{y})^2 / (n - 1)}, \quad \text{AIC} = c + n \ln(\sum e_i^2) + 2k.$$

**Logistic regression.**  $\text{logit}(p_i) = \log \frac{p_i}{1-p_i} = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}$ ,  $\text{OR}_j = e^{\beta_j}$ , sensitivity =  $\mathbb{P}(\text{Test+} | \text{Cond+})$ , specificity =  $\mathbb{P}(\text{Test-} | \text{Cond-})$ .

**Quantile hints (standard normal).**  $z_{0.70} = 0.524$ ,  $z_{0.80} = 0.842$ ,  $z_{0.90} = 1.282$ ,  $z_{0.95} = 1.645$ ,  $z_{0.975} = 1.960$ ,  $z_{0.99} = 2.326$ ,  $z_{0.995} = 2.576$ .